

IoT Weather Station using ESP32

Abstract

The IoT Weather Station is an ESP32-based environmental monitoring system that measures temperature, humidity, atmospheric pressure and altitude using DHT22 and BMP180 sensors. The ESP32 processes sensor data, predicts weather conditions based on atmospheric pressure and displays all readings on a 128×64 OLED display. An RGB LED visually indicates Sunny, Cloudy or Rainy conditions. The project demonstrates real-time sensing, data processing and embedded system integration.

Introduction

Weather monitoring is important in agriculture, environmental monitoring, research laboratories and smart city applications. This project provides a compact, low-cost weather station capable of monitoring environmental conditions continuously.

Objectives

- Measure temperature and humidity.
- Measure atmospheric pressure.
- Calculate altitude.
- Predict weather using pressure.
- Display information on OLED.
- Provide RGB LED indication.
- Print readings to the Serial Monitor.

Project Scope

The system is designed for educational and prototype purposes. It continuously acquires data from connected sensors, processes the readings in the ESP32 and updates the OLED display and RGB LED every second.

Hardware Components

ESP32 DevKit V1

DHT22 Temperature and Humidity Sensor

BMP180 Pressure Sensor

SSD1306 128×64 OLED Display

RGB LED

Jumper Wires

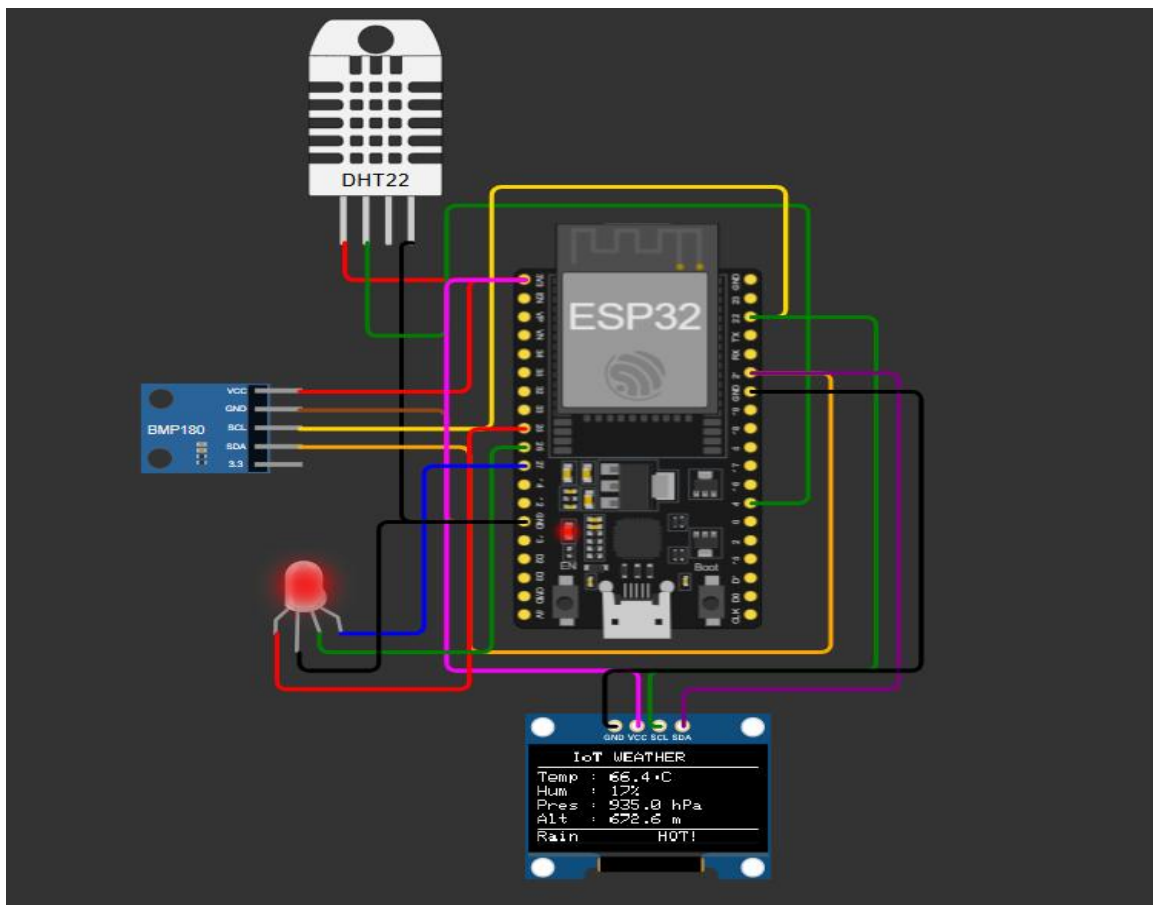
Software Requirements

Arduino IDE
ESP32 Board Package
Adafruit GFX Library
Adafruit SSD1306 Library
Adafruit BMP085 Library
DHT Sensor Library
Wokwi Simulator

Working Principle

After power-on, the ESP32 initializes the OLED and sensors. Every second, it reads temperature and humidity from the DHT22 and pressure and altitude from the BMP180. Atmospheric pressure is compared with predefined thresholds to estimate weather conditions. The RGB LED changes colour according to the prediction, while all values are displayed on the OLED and printed to the Serial Monitor.

Circuit Diagram



Algorithm

1. Start
2. Initialize ESP32 and peripherals
3. Read DHT22
4. Read BMP180
5. Calculate pressure and altitude
6. Predict weather
7. Update RGB LED
8. Update OLED
9. Print Serial data
10. Repeat

Weather Logic

Pressure >1015 hPa → Sunny (Green LED)

1005–1015 hPa → Cloudy (Yellow LED)

<1005 hPa → Rain (Red LED)

Advantages

Low cost, real-time monitoring, compact hardware, easy to expand, suitable for educational IoT projects.

Limitations

Weather prediction is based only on atmospheric pressure and is intended for demonstration purposes. The project does not upload data to the cloud.

Applications

Smart homes, smart agriculture, educational laboratories, environmental monitoring, weather observation, IoT learning.

Future Scope

Integrate Wi-Fi and cloud platforms, mobile app dashboard, data logging, rain sensor, wind speed sensor, GPS and weather history graphs.

Result

The system successfully measures environmental parameters, predicts weather conditions and displays the information in real time. The RGB LED provides a quick visual indication of the predicted weather.

Conclusion

The IoT Weather Station demonstrates how multiple sensors can be integrated with the ESP32 to build a reliable and low-cost monitoring system. The project can serve as a foundation for advanced IoT-based weather monitoring applications.

References

1. ESP32 Documentation
2. Arduino Documentation
3. Adafruit Libraries
4. DHT22 Datasheet
5. BMP180 Datasheet